MDFPF-Net: Multidomain Feature Perceptual Fusion Network for Hyperspectral Unmixing

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***Abstract*—Hyperspectral unmixing (HU) is crucial for retrieving subpixel information. However, existing deep learning (DL) methods typically rely on original spatial–spectral information and struggle to fuse the complementary advantages of multidomain features, resulting in endmember confusion and inaccurate abundance estimation. To address these challenges, a multidomain feature perceptual fusion network (MDFPF-Net) is proposed for HU. The network constructs a parallel feature extraction framework across the spatial, spectral, and frequency domains. Specifically, the spatial–spectral feature extraction (SSFE) module adopts a dual-branch design to capture distinct spatial and spectral attributes, while a frequency-domain multibranch attention (FMBA) module is introduced to selectively suppress redundant responses and enhance frequency components with stronger separability under the guidance of fine-grained texture information, yielding a high-fidelity representation of material distributions. To enhance cross-domain complementarity, a cross-domain dynamic perception (CDDP) module is designed. It uses adaptive context-aware convolutional kernels to achieve semantic alignment and collaborative optimization between both spatial–spectral and spatial–frequency features, strengthening endmember spectral discriminability and accurately characterizing abundance distribution discrepancies. Then, learnable parameters are used to adaptively integrate the dual-path outputs, producing deeply fused multidomain features. Finally, a semantic-enhanced abundance generator (SEAG) is embedded within a Transformer encoder–decoder architecture. Endmember-wise collaborative attention enhancement together with lightweight constraints is applied to further improve unmixing accuracy. Extensive experiments on synthetic and real-world datasets demonstrate that MDFPF-Net achieves superior performance, yielding lower root-mean-square error (RMSE) and spectral angle distance (SAD) than state-of-the-art methods in most cases. The source code is available at https://github.com/Nuist-HSI-Group/MDFPF-Net**

***Index Terms*—Deep learning (DL), dynamic perception, frequency-domain feature extraction, hyperspectral unmixing (HU), multidomain feature fusion.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are characterized by high spectral resolution and abundant, continuous spectral information [1]. Benefiting from both imaging and spectroscopy, HSIs play an important role in a wide range of remote sensing applications, including water quality monitoring [2], vegetation ecological assessment [3], mineral resource exploration [4], [5], and land-cover classification [6], [7]. However, limited spatial resolution of imaging sensors often leads to mixed pixels, where a single pixel may contain multiple materials. This long-standing issue hinders fine-grained interpretation of ground objects in practical scenarios, such as tree species classification [8] and target detection [9], [10]. Hyperspectral unmixing (HU) provides an effective solution by decomposing mixed pixels into a set of pure material components (endmembers) and estimating their corresponding proportions (abundances) [11].

In HU, the nonlinear mixing model (NLMM) accounts for radiative transfer effects and multiple scattering phenomena. Before reaching the sensor, incident photons may successively interact with multiple material components within the instantaneous field of view. Such multiple interactions and scattering processes result in a complex nonlinear combination of the spectral responses from different materials. In contrast, the linear mixing model (LMM) assumes that the observed spectrum of each pixel is a linear combination of the endmember spectra. This model is subject to three widely adopted physical constraints: the endmember nonnegativity constraint (ENC), the abundance nonnegativity constraint (ANC), and the abundance sum-to-one constraint (ASC). Due to its simplicity and computational efficiency, LMM has been widely used in HU.

LMM-based HU methods can be broadly categorized into traditional methods and deep learning (DL)-based methods. Traditional methods are typically grouped into three families: geometry-based, nonnegative matrix factorization (NMF)-based, and sparse regression-based approaches. Geometry-based methods exploit the geometric structure of hyperspectral data in the feature space for endmember extraction, such as N-FINDER [12], pixel purity index (PPI) [13], and vertex component analysis (VCA) [14], and these methods are often combined with fully constrained least squares sgsnmf: spatial group sparsity regularized nonnegative matrix factorization for hyperspectral unmixing (FCLS) [15] for abundance estimation. NMF-based methods formulate unmixing as constrained matrix factorization by leveraging nonnegativity and linear

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mixing properties, including MVC-NMF [16] and SGSNMF [17]. Sparse regression-based methods treat unmixing as a sparse optimization problem with respect to a spectral library under physical constraints, including locally collaborative sparse regression [18], [19] and sparse unmixing with spatial regularizations and low-rank constraints [20], [21].

In recent years, DL has been widely explored for HU due to its strong representation learning capability [22], [23]. Autoencoder (AE)-based approaches are among the most representative paradigms: the encoder maps high-dimensional spectral observations into low-dimensional abundances, while the decoder reconstructs the original image using the learned endmembers and abundances. Early AE-based methods were mostly pixel-level networks that relied on simple convolutions for feature extraction and used fully connected layers as encoders/decoders, which typically focused on a single type of feature. Representative methods include stacked nonnegative sparse autoencoders for robust hyperspectral unmixing (SNSA) [24], uDAS [25], deep autoencoder networks for hyperspectral unmixing (DAEN) [26], spectral-spatial hyperspectral unmixing using multitask learning (MTAEU) [27], and CyCU [28]. Although effective, such single-feature representations may not fully characterize the rich information contained in HSIs.

To better exploit the intrinsic spatial–spectral information, researchers have introduced convolutional neural networks (CNNs) into HU. Lu et al. [29] explored spatial–spectral collaborative information using CNNs. Zhao et al. [30] employed stacked 3D-CNN encoders to extract spatial–spectral features. MiSiC [31] introduced an AE-style CNN framework with a simplex-volume regularization. Spatial-spectral collaborative unmixing network for hyperspectral images (SSCU) [32] adopted a dual-stream convolutional architecture to enhance spatial and spectral feature utilization.

Nevertheless, CNNs mainly capture local patterns and may struggle to model long-range dependencies and adaptive multifeature fusion, which can limit further performance gains. Therefore, subsequent studies have combined CNNs with attention mechanisms, Transformer-based architectures, and Mamba-based state-space models to enhance global modeling and feature fusion. Hyperspectral unmixing with multi-scale convolution attention network (HUMSCAN) [33] incorporated attention modules for spatial and spectral feature enhancement. A2SAN [34] performed unmixing via an endmember-guided attention strategy. A spatial-spectral adaptive fusion network for hyperspectral unmixing with endmember variability (SSAF) [35] integrated spatial and channel attention with multiloss supervision. DeepTrans [36] introduced a Transformer to strengthen global correlations over CNN features. UnDAT [37] used a dual-branch Transformer to model spatial and spectral dependencies, while multiscale aggregation transformer network for hyperspectral unmixing (MAT) [38] leveraged a Transformer for adaptive multiscale feature fusion. UNMamba [39] employed cascaded spatial–spectral Mamba blocks to efficiently model long-range dependencies, while MBUNet [40] combined CNNs with bidirectional Mamba for spatial–spectral modeling. Although the above approaches significantly enhance global modeling capability, they still primarily operate within the original spatial–spectral domain and therefore fail to explicitly exploit complementary representations beyond this domain.

In this context, a small number of studies have begun to explore the role of frequency-domain features in HU. For example, Zhan et al. [41] performed a wavelet transform on hyperspectral data and constructed a wavelet-domain AE for unmixing. Liu et al. [42] proposed a frequency-domain unmixing approach based on the perturbed LMM (PLMM). Xu et al. [43] expanded multifrequency feature dimensions via wavelet transforms and employed a branched architecture to mine differentiated spatial–spectral information. These studies, to some extent, demonstrate that frequency-domain features can improve unmixing performance. However, most existing frequency-domain methods remain confined to shallow frequency representations or adopt simple concatenation-based fusion with the original spatial–spectral features, which fails to adequately explore deeper complementary relationships and effective cross-domain collaboration mechanisms.

A. Motivation

By systematically reviewing existing LMM-based DL methods for HU, two major bottlenecks can be identified. First, most existing methods mainly rely on original spatial–spectral features for modeling, which may easily lead to endmember confusion and inaccurate abundance estimation in complex highly mixed scenes. Meanwhile, frequency-domain features can provide complementary information from the perspectives of global structures and fine-grained textures, but they remain insufficiently explored in existing unmixing methods. Second, the fusion mechanism for multidomain features is still underdeveloped. Existing methods commonly adopt static fusion strategies, which lack the ability to dynamically model the complementary relationships among different features and may dilute informative cues. When spatial, spectral, and frequency-domain features are jointly involved, simple fusion strategies may fail to fully exploit their complementary advantages due to the different attributes of different domain features.

To address these limitations, this article aims to construct an unmixing network that can extract and deeply fuse spatial, spectral, and frequency-domain features in parallel. In recent years, frequency-domain processing techniques, such as wavelet and Fourier transforms, have been widely introduced into neural networks to enhance image structural representation and detail modeling capabilities [44], [45], [46]. Inspired by this, we design an attention-based frequency-domain feature extraction mechanism, in which multiple subband features obtained through wavelet decomposition are used to guide the adaptive refinement of global structural information extracted by the fast Fourier transform (FFT). This design suppresses interference while emphasizing detailed information valuable for the unmixing task, thereby enhancing the representation capability of land-cover distributions. The introduction of frequency-domain modeling is not intended to replace spatial–spectral features, but to complement their information representation. It should be noted that frequency-domain modeling cannot eliminate the original physical mixing effects. Instead, it provides complementary edge, texture, and fine-detail information, thereby enriching feature representations and facilitating subsequent

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multidomain fusion. Specifically, spatial features describe pixel neighborhood relationships, spectral features characterize spectral responses, and frequency-domain features represent land-cover texture details. The effective collaborative modeling of these three types of features can fully utilize the complementarity of different domain information, thereby improving endmember discrimination and abundance estimation accuracy. Furthermore, since simple fusion strategies are difficult to effectively model the dynamic complementary relationships among different domain features and may cause informative cues to be diluted or different features to compete with each other, we further draw on the ideas of multifeature fusion [47], [48] and adaptive fusion [49], [50] in the computer vision community. Motivated by the advantages of dynamic convolution in modeling contextual dependencies and local inductive biases [51], we design a cross-domain dynamic perceptual fusion mechanism. This mechanism separately models the complementary relationships between spatial–spectral and spatial–frequency features, and further enables the collaborative optimization of multidomain features. Overall, the proposed method provides a more effective multidomain feature representation and adaptive fusion scheme for HU.

B. Contribution

This article proposes a novel multidomain feature perceptual fusion network (MDFPF-Net) for HU. An end-to-end network is constructed within an AE architecture, in which stable frequency-domain information is introduced to complement the original spatial–spectral features. Unlike existing methods that mainly rely on spatial–spectral representations or simple aggregation of frequency-domain features, MDFPF-Net jointly models spatial, spectral, and frequency-domain information and performs progressive cross-domain interaction using spatial features as a common anchor. This design reduces interference among features from different domains, avoids information redundancy caused by simple feature stacking, and more fully exploits the complementarity of multidomain information. The main contributions are summarized as follows.

1) A parallel multidomain feature extraction framework is designed, where the spatial–spectral feature extraction (SSFE) extracts spatial and spectral features through a dual-branch convolutional structure. In addition, the frequency-domain multibranch attention (FMBA) generates multibranch attention scores guided by high-frequency texture details. It performs targeted enhancement of global structural information and conducts a weighted fusion of the multibranch outputs, yielding a stable frequency-domain representation of land-cover distribution for the unmixing process.
2) The cross-domain dynamic perception (CDDP) module produces feature-adaptive, context-aware convolution kernels to achieve semantic alignment and complementary enhancement of spatial–spectral and spatial–frequency features. These cross-domain features are subsequently refused via learnable parameters, enabling progressive fusion of multidomain features with greater comprehensiveness and stronger discriminability.
3) The proposed semantic-enhanced abundance generator (SEAG) module enhances semantic embeddings via endmember-wise attention to improve endmember separability. Furthermore, convolutional smoothing and a temperature-scaled Softmax function serve as lightweight constraints to refine endmember and abundance estimation.

The remainder of this article is organized as follows. Section II elaborates the architecture of the proposed MDFPF-Net. Section III presents the experimental settings and results. Section IV provides a detailed analysis and discussion. Finally, Section V concludes this article.

II. Proposed Method

This section introduces the proposed MDFPF-Net, which consists of three components.

1) Feature extraction, responsible for extracting multidomain features.
2) Perceptual fusion, undertaking the fusion of multidomain features with CDDP module.
3) Transformer encoder–decoder with SEAG, tasked with encoding the fused features and decoding them to generate the unmixing results with SEAG module. The overall architecture of the proposed MDFPF-Net is illustrated in Fig. 1. Sections II-A–II-E will conduct an in-depth analysis of these components as well as other details in the network architecture.

A. Problem Formulation

The HSI has spatial size $H \times W$ with B spectral bands and is denoted as $\boldsymbol{I} \in \mathbb{R}^{B\times H\times W}$, where $N = H \times W$ denotes the total number of pixels. The endmember matrix is defined as $\boldsymbol{E} \in \mathbb{R}^{B\times P}$, where P is the number of endmembers, and the corresponding abundance is $\boldsymbol{A} \in \mathbb{R}^{P\times H\times W}$.

In the LMM, each pixel spectrum is modeled as a linear combination of endmember spectra, where the combination weights correspond to the abundance fractions. Accordingly, the observed spectral reflectance can be expressed as

$$\boldsymbol{I} = \boldsymbol{E}\boldsymbol{A} + \boldsymbol{M} \tag{1}$$

where $\boldsymbol{M} \in \mathbb{R}^{B\times H\times W}$ represents the noise and disturbance term.

Commonly, the endmember matrix must be nonnegative and the abundance map must satisfy two physical constraints: 1) the ANC and 2) the ASC. We reshape the abundance tensor $\boldsymbol{A} \in \mathbb{R}^{P\times H\times W}$ into a matrix $\tilde{\boldsymbol{A}} \in \mathbb{R}^{P\times N}$ by vectorizing the spatial dimensions; these constraints can be formulated as follows:

$$\begin{cases} \tilde{\boldsymbol{A}} \geq \boldsymbol{0} \\ \mathbf{1}_P^\top \tilde{\boldsymbol{A}} = \mathbf{1}_N^\top \end{cases} \tag{2}$$

where $(\cdot)^\top$ denotes the transpose operation and $\mathbf{1}_P \in \mathbb{R}^P$ and $\mathbf{1}_N \in \mathbb{R}^N$ are all-ones column vectors.

B. Multidomain Feature Extraction

The feature extraction module is composed of two parallel branches: SSFE and FMBA. In SSFE, we use stacked blocks of 3×3 and 1×1 2-D convolutions for feature extraction, as shown in Fig. 2. Specifically, the 3×3 convolution has

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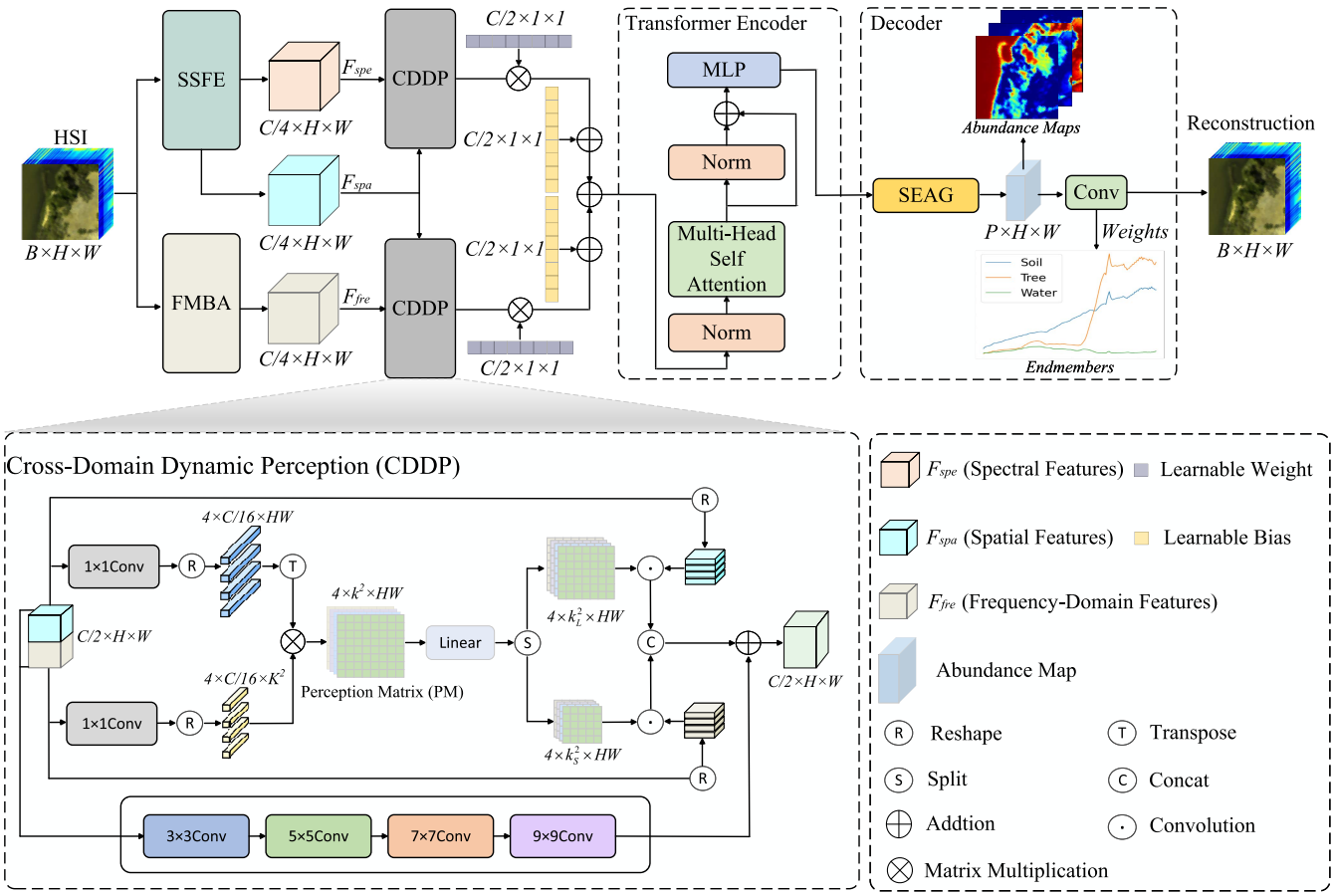


Fig. 1. Overall architecture of the proposed MDFPF-Net. The network extracts spatial–spectral and frequency-domain features, performs multidomain fusion, encodes the fused representation with a Transformer encoder, and finally estimates abundances and endmembers for hyperspectral reconstruction.

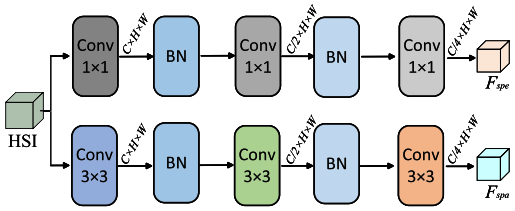


Fig. 2. Architecture of the SSFE module.

a moderate receptive field to directly capture local spatial correlation features of images. It accurately mines neighborhood dependencies between pixels and spatial structure information. The 1×1 convolution focuses on spectral dimension feature aggregation, enhancing information connections across different spectral bands. To ensure the stability and effectiveness of feature extraction, a batch normalization (BN) layer is added after each convolution. We also introduce the LeakyReLU activation function. It enriches feature expression through nonlinear transformation and effectively mitigates the risk of gradient explosion. The input HSI $\boldsymbol{I}$ is used to extract spatial features $\boldsymbol{F}_{spa} \in \mathbb{R}^{C/4\times H\times W}$ and spectral features $\boldsymbol{F}_{spe} \in \mathbb{R}^{C/4\times H\times W}$ through SSFE

$$\begin{cases} \boldsymbol{F}_{spa} = f_{31}\left(f_{21}\left(f_{11}\left(\boldsymbol{I}\right)\right)\right) \\ \boldsymbol{F}_{spe} = f_{32}\left(f_{22}\left(f_{12}\left(\boldsymbol{I}\right)\right)\right) \end{cases} \tag{3}$$

where $f_{ij}(\cdot)$ represents the combination of convolution, normalization, and activation functions with two different kernel sizes, and $C = 128$ is the number of feature channels.

Similarly, to extract stable frequency-domain attributes, FMBA integrates Fourier analysis and wavelet analysis, as shown in Fig. 3. Specifically, FMBA applies a band-wise FFT to the input HSI and obtains the global frequency representation $\boldsymbol{F}_{fft} \in \mathbb{R}^{B\times H\times W}$. In addition, FMBA employs the stationary wavelet transform (SWT) to decompose the input into one low-frequency approximation component and three directional high-frequency components, $\boldsymbol{LL}, \boldsymbol{LH}, \boldsymbol{HL}, \boldsymbol{HH} \in \mathbb{R}^{B\times H\times W}$, while keeping the spatial resolution unchanged. The corresponding mathematical formulation is given as follows:

$$\boldsymbol{F}_{fft}(u,v)_c = \left| \sum_{x=1}^{H} \sum_{y=1}^{W} \boldsymbol{I}(x,y)_c \; e^{-j2\pi\left(\frac{ux}{H}+\frac{vy}{W}\right)} \right| \tag{4}$$

$$\begin{cases} \boldsymbol{LL} = \boldsymbol{I} * (\psi_0 \otimes \psi_0) \\ \boldsymbol{LH} = \boldsymbol{I} * (\psi_0 \otimes \psi_1) \\ \boldsymbol{HL} = \boldsymbol{I} * (\psi_1 \otimes \psi_0) \\ \boldsymbol{HH} = \boldsymbol{I} * (\psi_1 \otimes \psi_1) \end{cases} \tag{5}$$

where $|\cdot|$ denotes the magnitude of the complex Fourier coefficient and j is the imaginary unit. u and v are the frequency indices. $\boldsymbol{I}(x,y)_c$ denotes the spatial value at location (x,y) of the cth band. Moreover, $*$ denotes the 2-D convolution

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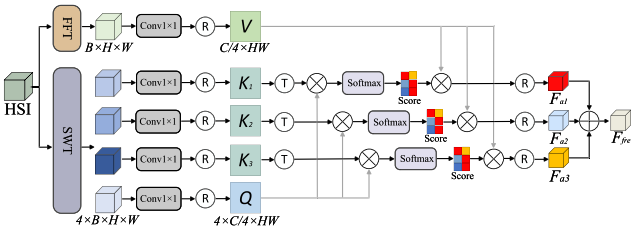


Fig. 3. Structure of the FMBA module.

operation; ψ_0 and ψ_1 are the low-pass and high-pass wavelet filters, respectively; and $\otimes$ denotes the tensor outer product.

The extracted $\boldsymbol{F}_{fft}$ is projected through a 1×1 convolution and subsequently reshaped to form the value matrix $\boldsymbol{V} \in \mathbb{R}^{4\times C/4\times HW} = \text{Reshape}(\boldsymbol{W}_v \boldsymbol{F}_{fft})$ in the attention mechanism. The low-frequency component $\boldsymbol{LL}$ is mapped to generate the query matrix $\boldsymbol{Q} \in \mathbb{R}^{4\times C/4\times HW} = \text{Reshape}(\boldsymbol{W}_q \boldsymbol{LL})$. Similarly, the three directional high-frequency subbands are individually transformed to produce the corresponding key matrix $\boldsymbol{K}_i \in \mathbb{R}^{4\times C/4\times HW} = \text{Reshape}(\boldsymbol{W}_{k_i} \boldsymbol{X}_i)$, $i \in \{1, 2, 3\}$, where $\boldsymbol{W}_v$, $\boldsymbol{W}_q$, and $\boldsymbol{W}_{k_i}$ denote 1×1 convolution operation, and Reshape$(\cdot)$ represents the reshape operation.

In order to ensure computational stability, FMBA applies global average pooling (GAP) along the spatial dimensions to $\boldsymbol{Q}$ and $\boldsymbol{K}_i$, transforming them into compact channel-wise descriptors. Subsequently, channel-domain attention is computed to emphasize structural information from different directions, and the resulting attention is further used to reweight the global frequency feature $\boldsymbol{V}$, producing the refined frequency features $\boldsymbol{F}_{ai}$ for three branches ($i = 1, 2, 3$). Finally, these refined features are adaptively fused to obtain the optimized frequency representation $\boldsymbol{F}_{fre}$. The specific calculation process is as follows:

$$\begin{cases} \boldsymbol{F}_{ai} = \text{Softmax}\left(\dfrac{\text{GAP}(\boldsymbol{Q}) \cdot \text{GAP}(\boldsymbol{K}_i)^T}{\sqrt{d}}\right) \cdot \boldsymbol{V} \\ \boldsymbol{F}_{fre} = \displaystyle\sum_{i=1}^{3} \gamma_i \boldsymbol{F}_{ai} \end{cases} \tag{6}$$

where GAP$(\cdot)$ denotes the GAP operation, $\boldsymbol{F}_{ai}$ represents the refined frequency feature of the ith branch, γ_i denotes the dynamic scaling coefficient for adaptive fusion, and d denotes the channel dimension of the projected query and key features ($d = C/4$), where $\sqrt{d}$ is used to scale the dot product for stable attention computation.

The FMBA is designed around frequency-domain feature extraction, where the multidirectional local frequency components obtained from wavelet decomposition are leveraged to guide the refinement of global frequency representations. This mechanism enables FMBA to adaptively focus on structurally informative frequency components while suppressing frequency bands that are more susceptible to noise. By transforming traditional spatial attention into lightweight channel-wise interactions, FMBA substantially reduces computational complexity while preserving strong representational capability.

C. Multidomain Perceptual Fusion With CDDP

After extracting multidomain features, directly aggregating them via simple concatenation or addition often leads to simple feature stacking, making it difficult to establish sufficient cross-domain interactions and complementary advantages, thereby limiting the unmixing performance. To address this issue, we propose a cross-domain dynamic perception CDDP module, as illustrated in Fig. 1 (bottom-left). During feature fusion, we employ two parallel CDDP modules to perceive the spatial–spectral information and the spatial–frequency information, respectively. Considering that spectral features mainly reflect material properties, while the frequency-domain features extracted from spatial transformations mainly characterize texture, boundary, and distribution details, we use spatial features as a common alignment anchor. Accordingly, two CDDP branches are adopted to separately model spatial–spectral and spatial–frequency complementarity, which avoids premature feature mixing and allows each branch to focus on a specific cross-domain relationship. Then, guided by learnable parameters, the cross-domain fused features generated by the two branches further interact and are fused again, yielding the final deep multidomain fused features.

Specifically, the CDDP first applies convolution, pooling, and reshaping operations to two types of cross-domain feature inputs $\boldsymbol{F}_1$ and $\boldsymbol{F}_2$, yielding the transformed features $\boldsymbol{F}'_1 \in \mathbb{R}^{4\times C/16\times HW}$ and $\boldsymbol{F}'_2 \in \mathbb{R}^{4\times C/16\times k^2}$. Subsequently, $\boldsymbol{F}'_1$ and $\boldsymbol{F}'_2$ are partitioned along the channel dimension into multiple mutually independent subspaces, enabling different subspaces to learn cross-domain relationships in parallel. This design not only improves computational efficiency but also effectively reduces the parameter scale and computational cost. Within each subspace, a perception matrix $\boldsymbol{PM} \in \mathbb{R}^{4\times k^2\times HW}$ is constructed via simple matrix multiplication to characterize the contextual correlations between features.

To generate context-aware convolutional kernels that are adaptive to features from different domains, CDDP employs a linear layer to enhance the discriminative semantic information of $\boldsymbol{PM}$ and uses a Softmax normalization to obtain the kernel weights. The resulting weights are then divided into two scales of context-aware convolutional kernels: a large-scale kernel $\boldsymbol{K}_L \in \mathbb{R}^{4\times k_L^2\times H\times W}$ and a small-scale kernel $\boldsymbol{K}_S \in \mathbb{R}^{4\times k_S^2\times H\times W}$. The large-scale kernel is mainly used for spatial-domain modeling, capturing long-range spatial dependencies and structured information of the target pixel and its neighborhood by leveraging a larger receptive field. In contrast, the small-scale kernel focuses on modeling spectral variations and responses to differences in land-cover distribution, ensuring accurate preservation of spectral and contour details. This design establishes a directed match between feature types and kernel scales: at each spatial location, the corresponding kernel is used to convolve the neighborhood features, enabling deep interactive aggregation across domains and fully exploiting the potential discriminative information, thereby effectively promoting semantic alignment and complementary benefits across domains. Finally, the processed feature groups are concatenated along the channel dimension and fed into the subsequent fusion module to further integrate multidomain representations.

For further preserving the structural integrity, we introduce a residual detail-compensation strategy with cascaded large receptive field convolutions. Given the cross-domain feature inputs, receptive fields at different scales collaboratively

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amplify local structural details, thereby strengthening the representation capability of fine-grained structures. Subsequently, the detail compensation produced by cascaded large receptive field convolutions is incorporated into the concatenated grouped features through a residual connection. This fusion process preserves the differential discriminability of the grouped features while incorporating multiscale local detail compensation from the convolutional residuals. Consequently, it produces spatial–spectral features $\boldsymbol{F}_{ss} \in \mathbb{R}^{C/2\times H\times W}$ that emphasize endmember spectral discriminability and spatial–frequency features $\boldsymbol{F}_{sf} \in \mathbb{R}^{C/2\times H\times W}$ that characterize differences in endmember distribution. The above procedure can be expressed as follows:

$$\boldsymbol{K}\odot\boldsymbol{F}=\boldsymbol{K}_{ij}*\boldsymbol{F}_{ij}, i\in\{1,\ldots,H\}, j\in\{1,\ldots,W\} \tag{7}$$

$$\begin{cases}\boldsymbol{F}_{ss}=\text{Concat}\left(\boldsymbol{K}_{L1}\odot\boldsymbol{F}_{spa};\boldsymbol{K}_{S1}\odot\boldsymbol{F}_{spe}\right)+\boldsymbol{R}_{ss}\\ \boldsymbol{F}_{sf}=\text{Concat}\left(\boldsymbol{K}_{L2}\odot\boldsymbol{F}_{spa};\boldsymbol{K}_{S2}\odot\boldsymbol{F}_{fre}\right)+\boldsymbol{R}_{sf}\end{cases} \tag{8}$$

where $\boldsymbol{F}$ denotes an arbitrary cross-domain feature and $\boldsymbol{K}$ denotes an arbitrary context-aware convolutional kernel. The operator $\odot$ denotes a dynamic convolution operation, where a position-dependent kernel $\boldsymbol{K}_{ij}$ is applied at each spatial location (i, j), Concat$(\cdot)$ represents the channel-wise concatenation operation, and $\boldsymbol{R}$ denotes the residual detail compensation produced by the cascaded large receptive field convolutions branch. The adaptive context-aware convolutional kernels $\{\boldsymbol{K}_{L1}, \boldsymbol{K}_{S1}\}$ and $\{\boldsymbol{K}_{L2}, \boldsymbol{K}_{S2}\}$ are generated by the spatial–spectral and spatial–frequency CDDP branches, respectively.

Finally, the spatial–spectral and spatial–frequency features are dynamically integrated by introducing learnable channel-wise weight and learnable bias parameters, so that complementary information across different domains can be fully exploited while redundant responses are suppressed, thereby achieving deep fusion of the three-domain representations. The resulting fused feature is then projected via a 1×1 convolution to match the channel dimension required by the subsequent Transformer encoder–decoder architecture, yielding the multidomain fused feature $\boldsymbol{F}_{ssf} \in \mathbb{R}^{8P\times H\times W}$, which can be formulated as follows:

$$\boldsymbol{F}_{ssf}=\text{Conv}_{1\times1}\left(\left(\boldsymbol{w}_1\cdot\boldsymbol{F}_{ss}+\boldsymbol{b}_1\right)+\left(\boldsymbol{w}_2\cdot\boldsymbol{F}_{sf}+\boldsymbol{b}_2\right)\right) \tag{9}$$

where $\boldsymbol{w}_i$ is the learnable channel-wise weight parameter and $\boldsymbol{b}_i$ is the learnable bias parameter.

D. Transformer Encoder–Decoder With SEAG

The Transformer encoder aims to transform the fused features into semantic coding. $\boldsymbol{F}_{ssf}$ is first flattened and partitioned into a sequence of tokens, after which the positional encodings $\boldsymbol{pos} \in \mathbb{R}^{L\times D}$ are added to preserve spatial ordering. A learnable semantic vector $\boldsymbol{c}_0 \in \mathbb{R}^{1\times D}$ is then prepended to the sequence, forming the initial input $\boldsymbol{Z} \in \mathbb{R}^{L'\times D}$

$$\boldsymbol{Z}=[\,\boldsymbol{c}_0,\,\boldsymbol{z}_1,\,\ldots,\,\boldsymbol{z}_L\,] \tag{10}$$

where $L = H/5 \times W/5$, $L' = L + 1$, $D = 200P$, and $\boldsymbol{z}_i$ denotes the ith feature token.

The encoder normalizes $\boldsymbol{Z}$ and feeds it into a multihead self-attention module. Unlike the standard Transformer, the query projection $\boldsymbol{w}_q$ is applied only to $\boldsymbol{c}_0$, while the key and value projections $\boldsymbol{w}_k$ and $\boldsymbol{w}_v$ are applied to the full sequence $\boldsymbol{Z}$. Then, the query, key, and value tensors are split into N_h heads, and each head independently computes scaled dot-product attention. The outputs of all heads are concatenated and then linearly projected to obtain the attention result, followed by a residual connection and layer normalization to stabilize the training. Subsequently, a multilayer perceptron (MLP) is applied to introduce nonlinear transformations, which enhances representational capacity. Finally, the global semantic coding $\boldsymbol{S} \in \mathbb{R}^{1\times D}$ is obtained by combining the updated semantic token $\boldsymbol{c}_0'$ with the mean of all feature tokens, expressed as

$$\boldsymbol{S}=\boldsymbol{c}_0'+\frac{1}{L}\sum_{i=1}^{L}\boldsymbol{z}_i'. \tag{11}$$

This coding integrates both global semantics and local dependency information, thereby providing subsequent decoding with more discriminative semantic information.

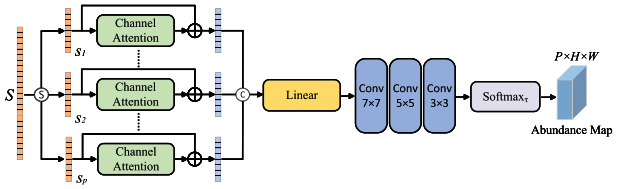


Fig. 4. Detailed architecture of the SEAG module.

Based on the semantic coding $\boldsymbol{S}$, the decoder further generates abundance maps and reconstructs the HSI. As shown in Fig. 4, $\boldsymbol{S}$ is first divided along the channel dimension into $\{\boldsymbol{s}_1, \boldsymbol{s}_2, \ldots, \boldsymbol{s}_P\}$. For each branch $\boldsymbol{s}_i$, SEAG applies channel attention to adaptively enhance the semantic features and employs a residual mapping to preserve the original semantic information, enabling the enhanced semantics to complement the original ones. Subsequently, the enhanced feature branches are concatenated and linearly projected into an intermediate abundance representation space, producing an initial abundance response $\boldsymbol{A}' \in \mathbb{R}^{P\times H\times W}$. To further improve spatial consistency and unmixing accuracy, SEAG introduces a cascaded convolution structure on this intermediate abundance, enforcing local consistency constraints and noise refinement, while simultaneously maintaining large-scale smoothness and recovering small-scale structural details, yielding a convolutionally smoothed abundance response $\boldsymbol{A}'' \in \mathbb{R}^{P\times H\times W}$. Subsequently, we employ a learnable temperature-scaled Softmax, denoted as Softmax$_\tau$, to normalize the refined responses $\boldsymbol{A}''$, so as to ensure that the generated abundance maps $\hat{\boldsymbol{A}} \in \mathbb{R}^{P\times H\times W}$ satisfy both ANC and ASC, thereby improving unmixing robustness. The corresponding formulation is given by

$$\boldsymbol{A}''=\text{Conv}_{7\times7}\left(\text{Conv}_{5\times5}\left(\text{Conv}_{3\times3}\left(\boldsymbol{A}'\right)\right)\right) \tag{12}$$

$$\hat{\boldsymbol{A}}_{phw}=\text{Softmax}_\tau\left(\boldsymbol{A}''_{phw}\right)=\frac{\exp\left(\boldsymbol{A}''_{phw}/\tau\right)}{\sum_{i=1}^{P}\exp\left(\boldsymbol{A}''_{ihw}/\tau\right)} \tag{13}$$

where $\hat{\boldsymbol{A}}_{phw} \in \hat{\boldsymbol{A}}$ denotes the abundance coefficient of the pth endmember at spatial location (h, w) and τ is a temperature

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Algorithm 1 MDFPF Network

Input:
Original HSI $\boldsymbol{I} \in \mathbb{R}^{B\times H\times W}$.

Output:
Estimated endmember $\hat{\boldsymbol{E}} \in \mathbb{R}^{B\times P}$, estimated abundance $\hat{\boldsymbol{A}} \in \mathbb{R}^{P\times H\times W}$, reconstructed HSI $\hat{\boldsymbol{I}} \in \mathbb{R}^{B\times H\times W}$.

1: Feed the original HSI $\boldsymbol{I}$ into the SSFE and FMBA modules to extract three domain features according to Eqs. (3)–(6): $\boldsymbol{F}_{spa}$ (spatial), $\boldsymbol{F}_{spe}$ (spectral), and $\boldsymbol{F}_{fre}$ (frequency).
2: Send $(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe})$ and $(\boldsymbol{F}_{spa}, \boldsymbol{F}_{fre})$ into the CDDP module to generate the spatial–spectral feature $\boldsymbol{F}_{ss}$ and the spatial–frequency feature $\boldsymbol{F}_{sf}$, respectively. Then, fuse them by Eq. (9) to obtain the spatial–spectral–frequency fused feature $\boldsymbol{F}_{ssf}$.
3: Convert the fused feature $\boldsymbol{F}_{ssf}$ into the semantic embedding $\boldsymbol{S}$ via Eqs. (10) and Eqs. (11)
4: Enhance the semantic embedding $\boldsymbol{S}$ using the SEAG module to predict the estimated abundance $\hat{\boldsymbol{A}} \in \mathbb{R}^{P\times H\times W}$.
5: Reconstruct the HSI using convolution to obtain $\hat{\boldsymbol{I}} \in \mathbb{R}^{B\times H\times W}$, and extract the convolutional weights as the estimated endmember $\hat{\boldsymbol{E}} \in \mathbb{R}^{B\times P}$.

coefficient optimized during training, which controls the sharpness of the abundance distribution.

Finally, a 1×1 convolution is applied to reconstruct the HSI $\hat{\boldsymbol{I}} \in \mathbb{R}^{B\times H\times W}$, where the convolution weights correspond to the endmember matrix $\hat{\boldsymbol{E}} \in \mathbb{R}^{B\times P}$. The initial weights of this layer are initialized using the endmembers extracted by VCA.

E. Loss Function

For model training, we adopted a composite loss function that integrates two distinct loss terms: reconstruction error (RE) loss and spectral angle distance (SAD) loss

$$L_{RE}\left(\boldsymbol{I}, \hat{\boldsymbol{I}}\right) = \frac{1}{B\times N}\sum_{b=1}^{B}\sum_{n=1}^{N}\left(\boldsymbol{I}_{bn} - \hat{\boldsymbol{I}}_{bn}\right)^2 \tag{14}$$

$$L_{\text{SAD}}\left(\boldsymbol{I}, \hat{\boldsymbol{I}}\right) = \frac{1}{N}\sum_{n=1}^{N}\arccos\left(\frac{\left\langle \boldsymbol{I}_n, \hat{\boldsymbol{I}}_n\right\rangle}{\|\boldsymbol{I}_n\|_2 \left\|\hat{\boldsymbol{I}}_n\right\|_2}\right) \tag{15}$$

where $\boldsymbol{I}_{bn}$ and $\hat{\boldsymbol{I}}_{bn}$ denote the spectral value at the bth band of the nth pixel in the original image $\boldsymbol{I}$ and the reconstructed image $\hat{\boldsymbol{I}}$, respectively. Moreover, $\boldsymbol{I}_n$ and $\hat{\boldsymbol{I}}_n$ are the spectral vectors of the nth pixel in $\boldsymbol{I}$ and $\hat{\boldsymbol{I}}$, respectively.

The total loss is a weighted combination of the two losses

$$L = \alpha L_{RE} + \beta L_{\text{SAD}} \tag{16}$$

where $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are hyperparameters.

F. Algorithm Summary for MDFPF-Net

The specific process of the proposed MDFPF-Net is shown in Algorithm 1.

III. Experiment

A. Dataset Description

Experiments are conducted on one synthetic HSI dataset and five real-world HSI datasets to thoroughly validate the effectiveness and generalization of the proposed MDFPF-Net. Fig. 5 shows the true-color images of these datasets together with their reference endmember.

1) Synthetic Dataset [23]: This dataset is a typical synthetic HU dataset, covering the wavelength range of 390–2470 nm, consisting of 224 bands and 105×105 spatial pixels. This dataset is synthesized by six endmembers based on the linear mixture model. The maximum pixel purity of this synthetic dataset is 0.75. Since there are no pure pixels in the scene and the pixel mixture degree is high, it can effectively represent the hyperspectral mixing characteristics under complex conditions. Therefore, it is often used to evaluate the robustness and accuracy of unmixing algorithms.

2) Samson Dataset [52]: This dataset is widely used in HU research. It has a spatial resolution of 952×952 pixels, with each pixel captured across 156 spectral bands. The wavelength range spans from 401 to 889 nm, and the spectral resolution reaches 3.13 nm. Due to the large size of the original image, which leads to high computational costs, a 95×95 pixel subset is typically used for unmixing experiments. The dataset consists of three distinct endmembers: soil, trees, and water.

3) Jasper Dataset [53]: Acquired by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor, this dataset is widely used in HU research. The original image has a size of 512×614 pixels with 224 spectral bands, covering wavelengths from 380 to 2500 nm. Due to the large dimensions of the full scene, a 100×100 pixel subimage was selected for this study. To mitigate atmospheric effects and exclude water absorption regions, bands 1–3, 108–112, 154–166, and 220–224 were discarded, leaving 198 bands for subsequent analysis. The resulting subimage, after removing bands affected by water vapor and atmospheric interference, contains 100×100 pixels with 198 bands and includes four endmembers: soil, water, trees, and roads.

4) Apex Dataset [36]: The dataset was acquired by the airborne prism experiment (APEX) sensor, with a spatial–spectral dimension of $1500 \times 1000 \times 285$ (spatial: 1500×1000 pixels; spectral: 285 bands) and a wavelength coverage of 413–2420 nm. After removing bands with low signal-to-noise ratio (SNR), a 110×110 pixel subimage was selected for the experiments. This subimage consists of four endmembers: road, trees, roof, and water.

5) Orchard Dataset [54], [55], [56]: This quasi-real hyperspectral scene was generated using an extended physically based ray-tracing (PBRT) model to evaluate nonlinear unmixing methods. The scene consists of 20×20 pixels with a spatial resolution of 2 m. It includes three endmember classes: tree, weed, and soil. The virtual sensor was initially configured with 216 spectral bands and a spectral resolution of 10 nm.

6) Cuprite Dataset [21]: Cuprite is a widely used real mineral dataset for HU, derived from the AVIRIS Cuprite mineral map. The original data contain 224 spectral bands. In the experiments, a 250×190 pixel subregion was selected

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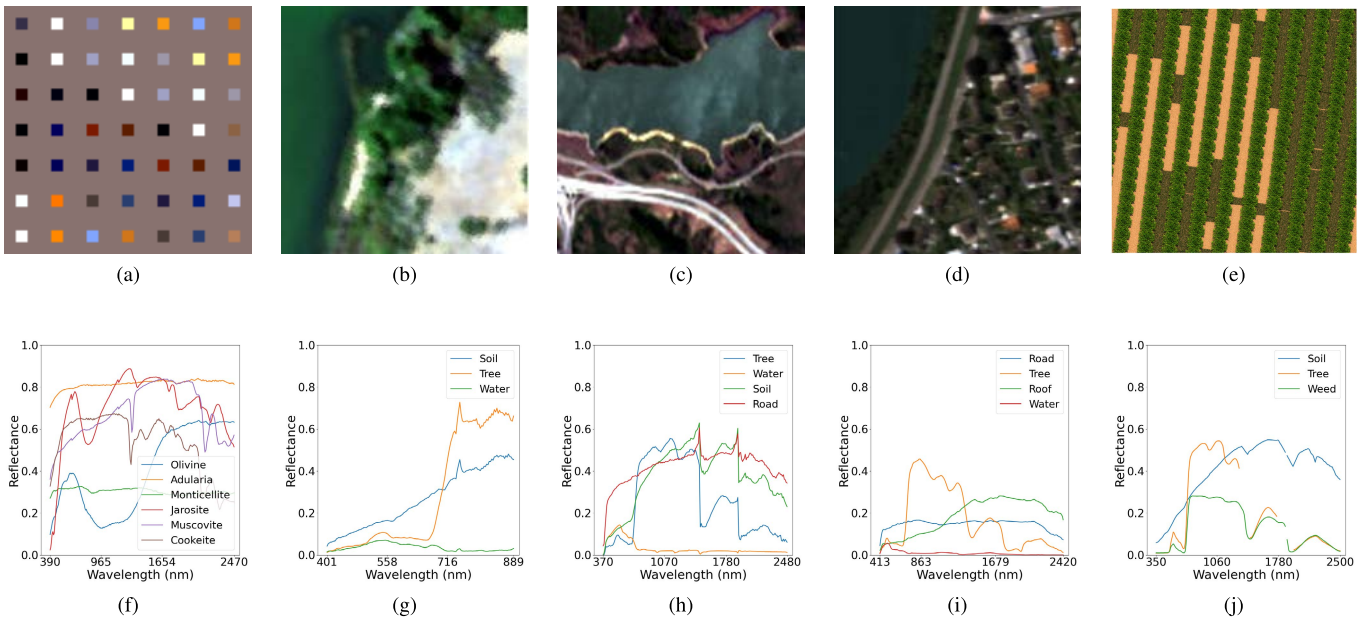


Fig. 5. True-color images and their reference endmembers of the experimental datasets. (a) Synthetic true-color image. (b) Samson true-color image. (c) Jasper Ridge true-color image. (d) Apex true-color image. (e) Orchard true-color image. (f) Synthetic reference endmembers (224 bands, 390–2470 nm). (g) Samson reference endmembers (156 bands, 401–889 nm). (h) Jasper Ridge reference endmembers (198 bands, 370–2457 nm). (i) Apex reference endmembers (285 bands, 413–2420 nm). (j) Orchard reference endmembers (216 bands, 350–2500 nm).

for analysis. Due to the effects of severe noise and water vapor absorption, bands 1, 2, 105–115, 150–170, 223, and 224 were removed, leaving 188 valid bands in the final dataset. This dataset contains abundant mineral distribution information and is commonly used to evaluate the performance of HU algorithms. The corresponding red, green, and blue (RGB) image and reference endmember spectra are provided in the *Supplementary Material Appendix C*.

B. Evaluation Metrics

The root-mean-square error (RMSE) is employed as a performance metric to quantify the estimation accuracy of abundance fractions by measuring the discrepancy between the reference abundance $\boldsymbol{A}$ and the estimated abundance $\hat{\boldsymbol{A}}$. In addition, the overall average RMSE (aRMSE) across all abundances is computed to provide a comprehensive evaluation. The corresponding mathematical formulations are defined as follows:

$$\mathrm{RMSE}\left(\boldsymbol{A},\hat{\boldsymbol{A}}\right)_p = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\left(\boldsymbol{a}_{pn}-\hat{\boldsymbol{a}}_{pn}\right)^2} \tag{17}$$

$$\mathrm{aRMSE} = \frac{1}{P}\sum_{p=1}^{P}\mathrm{RMSE}\left(\boldsymbol{A},\hat{\boldsymbol{A}}\right)_p \tag{18}$$

where a_{pn} denotes the reference abundance value of the pth endmember at the nth pixel and $\hat{a}_{pn}$ denotes the corresponding estimated abundance.

To evaluate the precision of endmember extraction, the SAD between the reference endmembers $\boldsymbol{E}$ and the estimated endmembers $\hat{\boldsymbol{E}}$ is utilized, along with the overall average SAD (aSAD) across all endmembers. These metrics are formally defined as follows:

$$\mathrm{SAD}\left(\boldsymbol{E},\hat{\boldsymbol{E}}\right)_p = \arccos\left(\frac{\boldsymbol{e}_p^{\top}\hat{\boldsymbol{e}}_p}{\left\|\boldsymbol{e}_p\right\|_2\cdot\left\|\hat{\boldsymbol{e}}_p\right\|_2}\right) \tag{19}$$

$$\mathrm{aSAD} = \frac{1}{P}\sum_{p=1}^{P}\mathrm{SAD}\left(\boldsymbol{E},\hat{\boldsymbol{E}}\right)_p \tag{20}$$

where e_p denotes the reference endmembers vector of the pth endmember and $\hat{e}_p$ denotes the estimated endmember vector of the pth endmember.

C. Experimental Setup

1) Comparison Algorithms: In order to evaluate the effectiveness of this method, we selected ten representative or most advanced unmixing methods for comparative analysis. These methods cover different technical categories: common traditional methods, including VCA [14] and FCLS [15], which are the most classic combination of "endmember extraction and abundance estimation" in hyperspectral traditional unmixing, as well as space group sparse regular unmixing method SGSNMF [17] based on NMF. The advanced DL-based unmixing methods include: DAEN [26], CyCU [28], MiSiC [31], DeepTrans [36], A2SAN [34], MAT [38], SSAF [35], and UNMamba [39]. DAEN introduced the variational AE into HU for the first time and solved the problems of outliers and abundance constraints. CyCU focused on spatial information extraction of HSIs. MiSiC used both spatial and geometric features. DeepTrans was the first unmixing network based on Transformer architecture. A2SAN guided the Transformer for unmixing through endmembers. MAT focused on multiscale feature aggregation Transformer. SSAF integrated spatial and channel attention for effective feature fusion. UNMamba employed cascaded spatial–spectral Mamba modules to enhance feature representation. The initial endmembers of all comparison methods are extracted by the VCA algorithm.

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TABLE I
HYPERPARAMETER SETTINGS OF THE COMPARED METHODS

Method	Parameter	Synthetic	Samson	Jasper Ridge
VCA–FCLS [14]	–	–	–	–
SGSNMF [17]	MaxIter	200	100	100
DAEN [26]	MaxIter	500	500	500
CyCU [28]	LR	3e-3	2e-3	8e-3
	Epoch	600	500	300
MiSiC [31]	LR	1e-3	1e-3	1e-3
	Epoch	800	600	700
DeepTrans [36]	LR	4e-3	6e-3	6e-3
	Epoch	1000	200	200
	Weight decay	5e-5	4e-5	4e-5
A2SAN [34]	LR	5e-4	1e-3	5e-3
	Epoch	500	1000	300
	Weight decay	1e-4	1e-4	1e-4
MAT [38]	LR	2e-3	9e-3	9e-3
	Epoch	400	200	200
	Weight decay	5e-5	4e-5	4e-5
SSAF [35]	LR	5e-4	1e-3	5e-3
	Epoch	2000	2000	2000
UNMamba [39]	LR	4e-3	3e-3	3e-3
	Epoch	1000	800	1000
MDFPF-Net	LR	2e-3	2e-3	3e-3
	Epoch	400	100	100
	Weight decay	3e-5	4e-5	4e-5
	η	0.9	0.7	0.7
	N_h	8	8	8

2) Hyperparameter Settings: The hyperparameter settings of MDFPF-Net are summarized as follows. The momentum parameter of the BN layers is set to 0.9. The convolutional weights in the decoder are constrained within the range $(1 \times 10^{-5}, 1)$ to satisfy physical constraints. The initial temperature τ of Softmax_τ is set to 1. The network parameters are optimized using the Adam optimizer, with different initial learning rates (LRs) adopted for different datasets. To ensure stable convergence during training, the LR is decayed by a factor of η every 20 epochs using a step-based scheduling strategy. Weight decay is incorporated into the optimizer to alleviate overfitting, and all convolutional layers are initialized using the Kaiming initialization method. The dataset-specific hyperparameter settings are summarized in Table I. To ensure a fair and transparent comparison, we first adopted the configurations reported in the original papers or official implementations whenever dataset-specific settings were available. For example, the parameters of MAT and UNMamba on the Jasper Ridge dataset were directly set according to their respective published configurations. When no dataset-specific settings were provided, such as for SSAF on the Samson dataset, only the principal training-related hyperparameters, including the LR, batch size, and weight decay, were adjusted within reasonable ranges. The final settings were selected according to the convergence behavior and training stability of each method, with the aim of achieving satisfactory and consistently stable performance. This strategy was applied uniformly to all comparison methods.

D. Experimental Results

1) Synthetic Dataset: To evaluate the noise resistance and robustness of the proposed method, additive white Gaussian noise with different power levels was introduced into the synthetic dataset to generate observations with SNRs of 20, 30, and 40 dB. These noisy data were used to systematically assess the unmixing performance of different methods under various noise interference conditions.

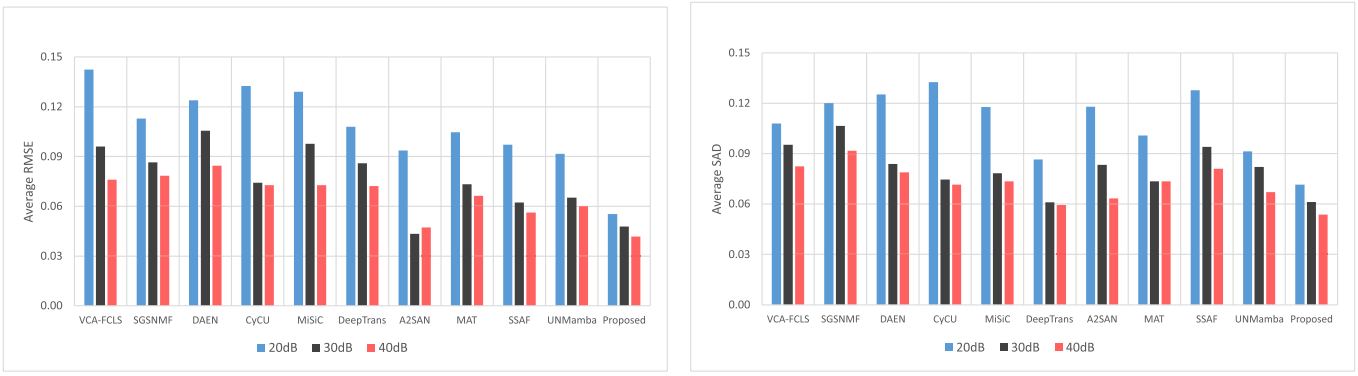
As shown in Fig. 6, the proposed method achieves the best aSAD performance under all three SNR conditions, while also maintaining excellent results in terms of aRMSE. Although the aRMSE of the proposed method is slightly inferior to that of A2SAN under the 30-dB condition, it still ranks as the second-best among the compared methods, and its aSAD is clearly better than that of A2SAN. This indicates that MDFPF-Net can more accurately recover the spectral shapes of endmembers while maintaining good abundance estimation accuracy. Furthermore, compared with SSAF, the proposed method achieves better aSAD performance under different noise conditions and maintains stable advantages in aRMSE. Overall, by introducing frequency-domain features and cross-domain dynamic perception fusion, MDFPF-Net effectively enhances the complementary modeling capability among multidomain information, thereby achieving more stable and reliable unmixing results under different noise conditions.

Furthermore, to more intuitively demonstrate the unmixing performance of the model in noisy environments, Figs. 7 and 8 present the abundance estimation results and endmember reconstruction results under the 20-dB condition, respectively. As shown in Fig. 7, the spatial distributions predicted by the proposed method are more consistent with the ground-truth abundance maps, with clearer target structures and less background noise. As shown in Fig. 8, the endmember spectra estimated by the proposed method are more consistent with the ground-truth spectra in terms of both the overall trend and spectral details, indicating that the proposed method can still accurately recover endmember and abundance information under noise interference. These results further verify the effectiveness and robustness of the proposed method in noisy scenarios.

2) Samson Dataset: Table II presents the quantitative evaluation results of all methods on the Samson dataset. Compared with other methods, the proposed method achieves the best overall performance in both aRMSE and aSAD. Most individual metrics also rank first or second among the compared methods, demonstrating strong overall abundance and endmember estimation capability.

Fig. 9 shows the abundance estimation results of all methods. It can be observed that the abundance maps generated

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(a) (b)

Fig. 6. Experimental results on synthetic dataset with different SNRs. (a) aRMSE. (b) aSAD.

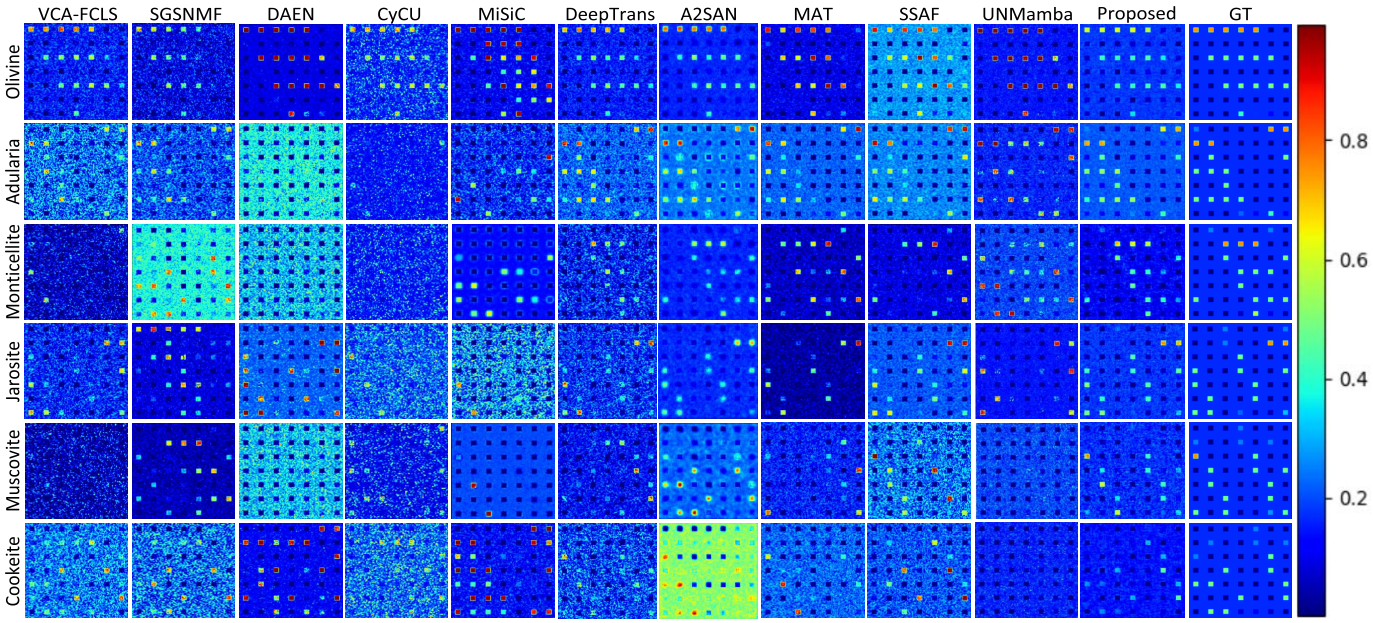


Fig. 7. Visual comparison of the abundances obtained by different methods on the synthetic dataset with an SNR of 20 dB.

TABLE II

QUANTITATIVE COMPARISON OF UNMIXING EFFECTS OF ALL EXAMINED METHODS ON SAMSON DATASET. RED: BEST RESULT AND BLUE: SECOND-BEST RESULT

Method		VCA-FCLS [14]	SGSNMF [17]	DAEN [26]	CyCU [28]	MiSiC [31]	DeepTrans [36]	A2SAN [34]	MAT [38]	SSAF [35]	UNMamba [39]	Proposed
RMSE	Tree	0.1588	0.1486	0.2186	0.0965	0.1243	0.0852	0.0722	0.1714	0.0462	0.1328	0.0435
	Water	0.0772	0.2549	0.1973	0.1478	0.0962	0.0715	0.0381	0.1325	0.0862	0.1750	0.0671
	Soil	0.1758	0.3351	0.3153	0.1712	0.0704	0.0982	0.2563	0.2369	0.0532	0.2583	0.0416
aRMSE		0.1373	0.2462	0.2437	0.1385	0.0970	0.0850	0.1222	0.1803	0.0619	0.1887	0.0507
SAD	Tree	0.0259	0.0443	0.0697	0.0902	0.0294	0.0374	0.0544	0.0165	0.0310	0.0238	0.0113
	Water	0.0958	0.0990	0.1036	0.0624	0.0787	0.0708	0.0900	0.0914	0.0793	0.0674	0.0684
	Soil	0.1554	0.0394	0.1891	0.4635	0.2956	0.0793	0.0359	0.0393	0.1744	0.0435	0.0381
aSAD		0.0924	0.0609	0.1208	0.2054	0.1346	0.0625	0.0601	0.0490	0.0949	0.0449	0.0393

by the proposed method exhibit the highest consistency with the reference abundances among all methods. This superiority stems from the model's in-depth mining and fusion of spatial, spectral, and frequency-domain features, which enables the use of multiple features to regulate and guide the generation of abundance maps. Due to their utilization of insufficiently comprehensive features, CyCU, DeepTrans, A2SAN, and MAT show a significant performance gap compared to the proposed method. In particular, they tend to confuse tree and soil types, misclassifying them in some pixels, which directly impairs the accuracy of the final abundance estimation results. Although SSAF achieves promising results, its depiction of

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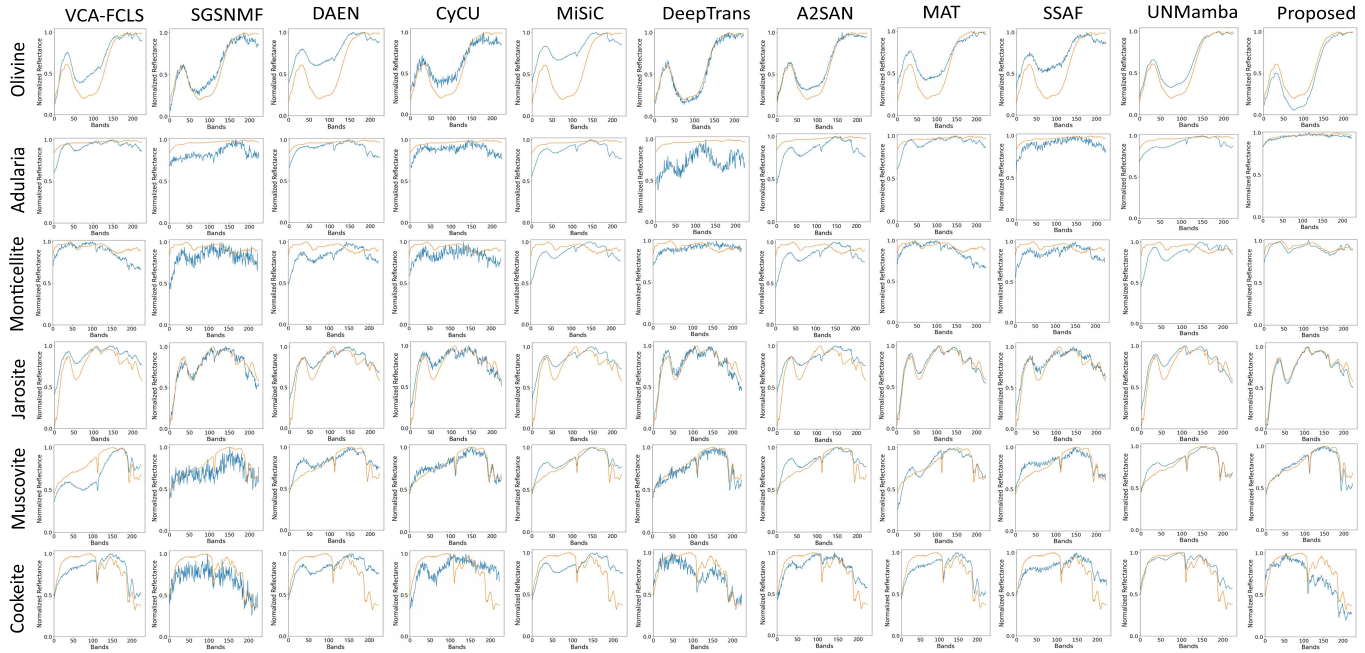


Fig. 8. Visual comparison of the endmembers obtained by different methods on the synthetic dataset with an SNR of 20 dB. Blue line: estimated endmembers and yellow line: reference endmembers.

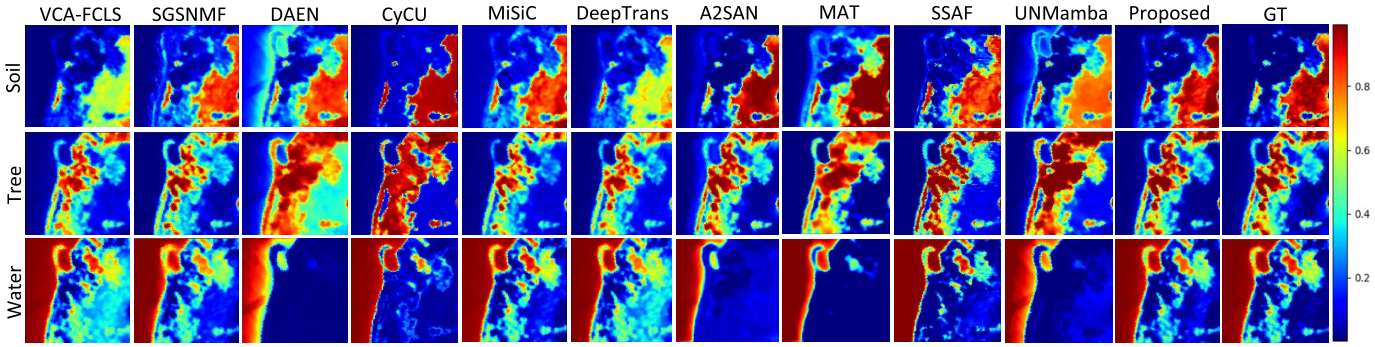


Fig. 9. Visual comparison of the abundances obtained by different methods on the Samson dataset.

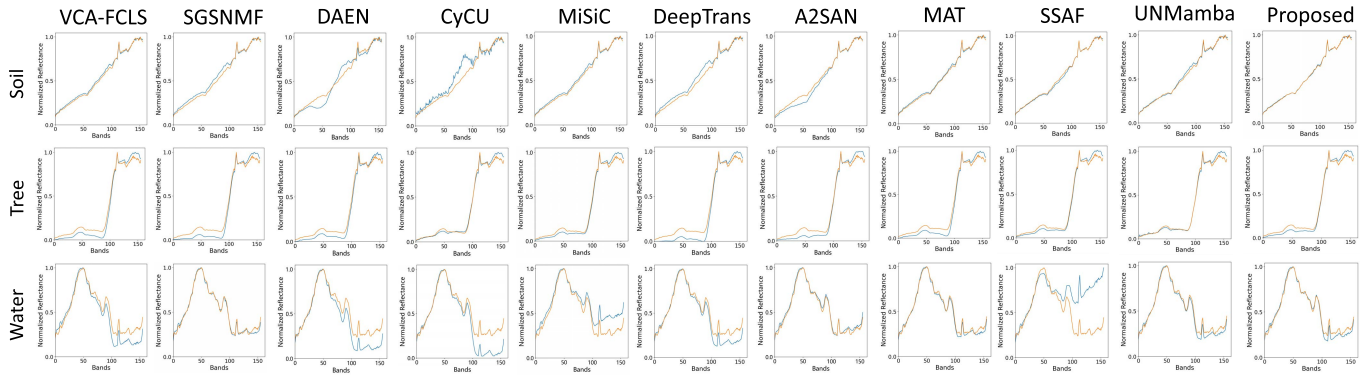


Fig. 10. Visual comparison of the endmembers obtained by different methods on the Samson dataset. Blue line: estimated endmembers and yellow line: reference endmembers.

certain details still needs improvement compared to the proposed method.

Fig. 10 illustrates the endmember estimation results of the compared methods. The proposed method achieves the best SAD for tree and the second-best result for soil while maintaining competitive performance for water. More importantly, it obtains the lowest overall aSAD among all compared methods.

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TABLE III

QUANTITATIVE COMPARISON OF UNMIXING EFFECTS OF ALL EXAMINED METHODS ON JASPER RIDGE DATASET. RED: BEST RESULT AND BLUE: SECOND-BEST RESULT

Method		VCA-FCLS [14]	SGSNMF [17]	DAEN [26]	CyCU [28]	MiSiC [31]	DeepTrans [36]	A2SAN [34]	MAT [38]	SSAF [35]	UNMamba [39]	Proposed
RMSE	Tree	0.1327	0.0819	0.1122	0.1010	0.1591	0.0882	0.0782	0.1820	0.0762	0.0581	0.0745
	Water	0.1468	0.1122	0.1684	0.4334	0.1214	0.1215	0.0503	0.0608	0.0962	0.0529	0.0491
	Soil	0.1509	0.1625	0.1498	0.1209	0.1839	0.1982	0.1145	0.1667	0.0732	0.1013	0.0921
	Road	0.1803	0.2048	0.1547	0.0709	0.2117	0.2452	0.0866	0.1871	0.0693	0.0869	0.0687
aRMSE		0.1527	0.1403	0.1463	0.1816	0.1690	0.1632	0.0824	0.1492	0.0787	0.0748	0.0711
SAD	Tree	0.1657	0.1035	0.1164	0.1729	0.1747	0.0974	0.0498	0.0416	0.0310	0.0593	0.0276
	Water	0.5459	0.1955	0.1577	0.3907	0.4099	0.1008	0.1393	0.0803	0.1893	0.0910	0.0332
	Soil	0.2234	0.0296	0.1425	0.0784	0.0887	0.1293	0.0597	0.4419	0.0344	0.0720	0.0271
	Road	0.0901	0.3130	0.1619	0.0589	0.1133	0.1550	0.1362	0.1688	0.0437	0.0495	0.1278
aSAD		0.2563	0.1604	0.1446	0.1752	0.1967	0.1206	0.0963	0.1832	0.0746	0.0680	0.0539

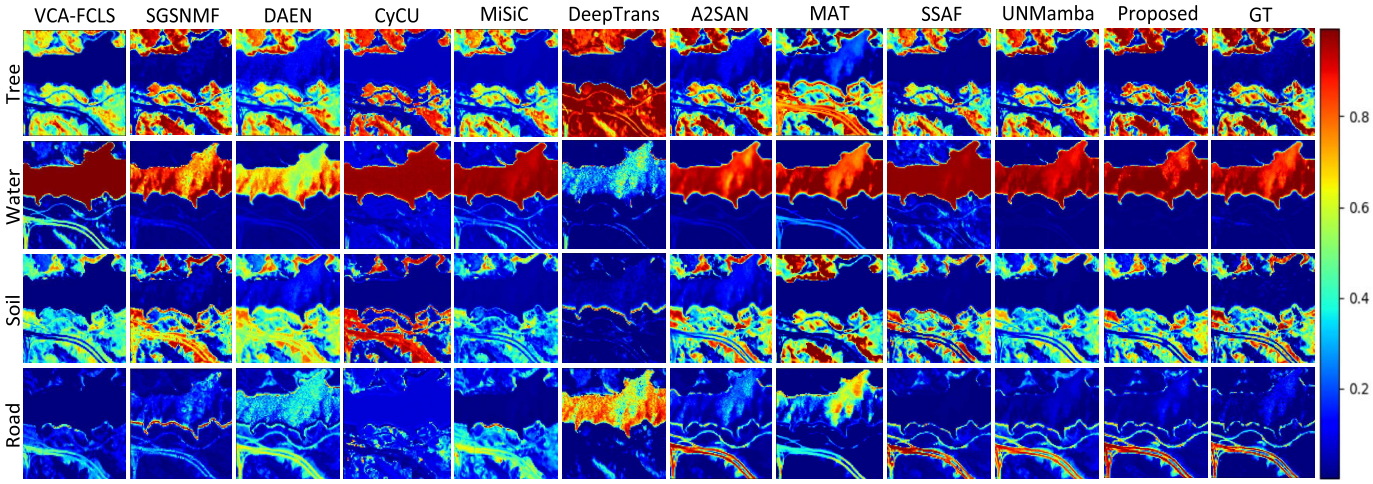


Fig. 11. Visual comparison of the abundances obtained by different methods on the Jasper Ridge dataset.

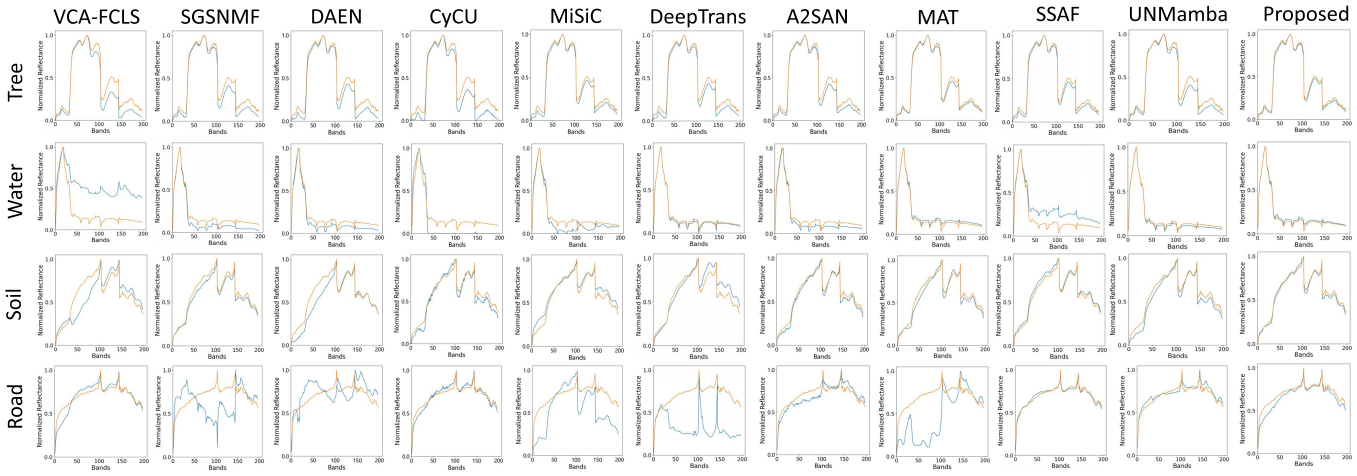


Fig. 12. Visual comparison of the endmembers obtained by different methods on the Jasper Ridge dataset. Blue line: estimated endmembers and yellow line: reference endmembers.

3) Jasper Ridge Dataset: Table III provides a detailed record of the quantitative evaluation results of all methods on the Jasper Ridge dataset. Compared with other methods, the proposed method exhibits extremely strong abundance and endmember estimation capabilities for the Water class, while achieving the best aRMSE and aSAD. It is evident that our method performs excellently overall.

Fig. 11 presents the abundance estimation results of all methods, from which it can be intuitively observed that the estimated abundances of the proposed method are the

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closest to the reference abundances among all approaches. The CyCU method lacks sufficient detail in depicting the texture and contour of the road class, leading to an overestimated abundance of the soil class in the road regions. DeepTrans tends to misclassify both the soil and road classes as the tree class. Methods such as VCA-FCLS, CyCU, MiSiC, and SSAF have limitations in estimating the abundance of the water class—they struggle to accurately distinguish the abundance differences across different water regions, resulting in a nearly uniform distribution in the water abundance map that fails to effectively reflect the true abundance distribution of water bodies.

Fig. 12 illustrates the endmember estimation results of all methods. It can be seen that the endmember curves estimated by the proposed method for the tree, water, and soil classes almost perfectly match the reference endmember curves. Although the estimated endmember curve for the road class is slightly inferior to other methods, its overall performance is acceptable when combined with its excellent performance in the abundance map.

Detailed quantitative and visual results for the Apex, Orchard, and Cuprite datasets are provided in *Appendixes A–C* of the Supplementary Material. On the Apex dataset, MDFPF-Net can effectively recover the main spatial distributions of different materials, and the estimated endmember spectra are generally consistent with the corresponding reference spectra. The Orchard dataset is more difficult to unmix due to multiple-scattering effects. Nevertheless, MDFPF-Net can still accurately estimate the spatial distributions of materials, and the estimated spectra are relatively close to the reference spectra. Although the estimation of the Tree endmember shows some deviation, the proposed method still achieves the best average evaluation metrics, demonstrating superior overall unmixing performance. On the Cuprite dataset, the estimated abundance maps can capture the main distribution regions of representative minerals, and the extracted spectra are also relatively close to the reference mineral spectra. These visual results further demonstrate the generalization capability of MDFPF-Net across different scenarios.

IV. Discussion

To verify the effectiveness of the proposed design, we conducted ablation experiments, including component-level analysis and an additional study on the volume loss. We also evaluated the running time across all datasets.

A. Ablation Study on Network Components

To evaluate the contribution of each component, we constructed several ablation variants based on the complete model, which consists of SSFE, FMBA, CDDP, and SEAG. Variant 1 removes SEAG. Variant 2 removes FMBA and retains only the spatial–spectral CDDP branch, while Variant 3 directly introduces $\boldsymbol{F}_{fre}$ into the subsequent fusion stage without dedicated dynamic modeling. Variant 4 replaces the original spatial–frequency branch with a spectral–frequency branch. Variants 5 and 6 replace CDDP with direct concatenation-based fusion and attention-based tri-domain fusion, respectively. Variant 7 first aggregates the three domain features into $\boldsymbol{F}_{tri} = \mathrm{Conv}_{1\times1}(\mathrm{Concat}(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe}, \boldsymbol{F}_{fre}))$ and then feeds $(\boldsymbol{F}_{tri}, \boldsymbol{F}_{tri})$ into a single CDDP. Variant 8 adds an additional spectral–frequency branch to form a three-branch CDDP structure. Variant 9 removes SSFE and adopts a single $(\boldsymbol{F}_{fre}, \boldsymbol{F}_{fre})$ branch. For the internal components of SEAG, Variant 10 replaces the learnable temperature-scaled Softmax with standard Softmax by fixing $\tau = 1$. Variant 11 removes the cascaded 3×3, 5×5, and 7×7 convolutions and directly applies the temperature-scaled Softmax to the initial abundance responses. Variant 12 removes the endmember-wise channel attention while retaining the linear projection, cascaded convolutions, and temperature-scaled Softmax.

As shown in Table IV, the complete model (Variant 0) achieves the best results on both metrics, indicating that the proposed modules work well together. In Variant 1, after removing SEAG, the performance drops noticeably, showing that this module is beneficial for abundance estimation and also helps improve endmember extraction. In Variant 2, after removing FMBA and retaining only the $(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe})$ branch, both aRMSE and aSAD increase significantly, which confirms that frequency-domain features provide useful complementary information for HU. In Variant 3, when $\boldsymbol{F}_{fre}$ is directly introduced into the subsequent fusion stage without dedicated cross-domain dynamic modeling, the performance is still clearly worse than that of the complete model. This indicates that simply appending frequency-domain information is insufficient and that its complementary value needs to be exploited through targeted cross-domain dynamic modeling. Similarly, although Variant 4, with the alternative dual-branch configuration $(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe})$ Branch $+(\boldsymbol{F}_{spe}, \boldsymbol{F}_{fre})$ Branch, also introduces dual-path fusion, its performance remains slightly lower than that of the original dual-branch design. This suggests that emphasizing spectral–frequency fusion may not effectively reflect fine-grained distribution details of ground objects. For the fusion strategy itself, both Variant 5, which adopts the simple fusion strategy $\mathrm{Conv}_{1\times1}(\mathrm{Concat}(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe}, \boldsymbol{F}_{fre}))$, and Variant 6, which adopts the attention-based tri-domain fusion strategy $\mathrm{AttnFusion}(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe}, \boldsymbol{F}_{fre})$, perform worse than the complete model. Although the attention-based fusion in Variant 6 brings certain improvement over the direct concatenation in Variant 5, it still cannot achieve the performance of the complete dual-CDDP structure. This indicates that simple feature fusion or ordinary attention-based reweighting is insufficient to fully model the complementary relationships among multidomain features, while the proposed CDDP can more effectively achieve dynamic cross-domain perception and complementary enhancement. More importantly, Variant 7, namely the single-CDDP tri-domain fusion variant denoted as the $(\boldsymbol{F}_{tri}, \boldsymbol{F}_{tri})$ Branch, still performs worse than the proposed dual-branch progressive fusion strategy on both metrics. This indicates that compressing the three types of features into a shared fusion space at once may help recover the overall abundance distribution to some extent, but is insufficient to simultaneously preserve the spectral shape consistency of endmembers. A possible reason is that spatial, spectral, and frequency-domain features have different attributes. When they are fused too early in a single branch, spectral discriminative information and frequency-domain detail information may

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TABLE IV
ABLATION STUDY RESULTS OF COMPONENTS. RED: BEST RESULT AND BLUE: SECOND-BEST RESULT

Variant No.	Component				Evaluation metrics	
	SSFE	FMBA	Fusion Configuration	SEAG Configuration	aRMSE	aSAD
0	✓	✓	Original dual-branch($(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe})$ Branch + $(\boldsymbol{F}_{spa}, \boldsymbol{F}_{fre})$ Branch)	✓	0.0594	0.0574
1	✓	✓	Original dual-branch	✗	0.0751	0.0834
2	✓	✗	$(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe})$ Branch without $(\boldsymbol{F}_{spa}, \boldsymbol{F}_{fre})$ Branch	✓	0.0842	0.0876
3	✓	✓	$(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe})$ Branch + $\boldsymbol{F}_{fre}$	✓	0.0803	0.0812
4	✓	✓	$(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe})$ Branch + $(\boldsymbol{F}_{spe}, \boldsymbol{F}_{fre})$ Branch	✓	0.0767	0.0761
5	✓	✓	$\mathrm{Conv}_{1\times1}(\mathrm{Concat}(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe}, \boldsymbol{F}_{fre}))$	✓	0.0868	0.0979
6	✓	✓	$\mathrm{AttnFusion}(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe}, \boldsymbol{F}_{fre})$	✓	0.0834	0.0917
7	✓	✓	$(\boldsymbol{F}_{tri}, \boldsymbol{F}_{tri})$ Branch, where $\boldsymbol{F}_{tri} = \mathrm{Conv}_{1\times1}(\mathrm{Concat}(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe}, \boldsymbol{F}_{fre}))$	✓	0.0731	0.1023
8	✓	✓	$(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe})$ Branch + $(\boldsymbol{F}_{spa}, \boldsymbol{F}_{fre})$ Branch + $(\boldsymbol{F}_{spe}, \boldsymbol{F}_{fre})$ Branch	✓	0.0819	0.0715
9	✗	✓	$(\boldsymbol{F}_{fre}, \boldsymbol{F}_{fre})$ Branch without $(\boldsymbol{F}_{spa}, \boldsymbol{F}_{spe})$ Branch	✓	0.0854	0.1016
10	✓	✓	Original dual-branch	Standard Softmax(τ=1)	0.0719	0.0763
11	✓	✓	Original dual-branch	Without Cascaded Convolution	0.0754	0.0852
12	✓	✓	Original dual-branch	Without Channel Attention	0.0675	0.0806

interfere with each other, thereby weakening the targeted modeling of specific cross-domain complementary relationships. In contrast, the proposed dual-branch progressive fusion strategy models spatial–spectral and spatial–frequency interactions separately and then integrates them, which is more effective for preserving endmember characteristics and abundance spatial distributions. Moreover, Variant 8, which adopts the three-branch CDDP structure, obtains the second-best aSAD but shows a clear degradation in aRMSE compared with the complete model, indicating that the additional spectral–frequency branch may help preserve spectral shape consistency to some extent, but can also disturb abundance estimation. In addition, Variant 9 removes SSFE and leads to a further performance drop, indicating that spatial and spectral feature extraction provides an important basis for the subsequent cross-domain fusion.

Variants 10–12 further evaluate the internal components of SEAG. Variant 10 performs worse than the complete model on both metrics, indicating that the learnable temperature coefficient can adaptively control the sharpness of the abundance distribution and improve abundance normalization. Variant 11 also leads to noticeable performance degradation, demonstrating that the cascaded convolutions play an important role in refining abundance structures. Variant 12 increases both error metrics, especially aSAD, indicating that endmember-wise channel attention helps enhance semantic information and improve the spectral discriminability of endmembers. Overall, the ablation study not only verifies the effectiveness of each component but also demonstrates the rationality of the proposed dual-branch progressive multidomain fusion architecture.

B. Ablation Study on Loss Function

It should first be clarified that MDFPF-Net is not a pure-pixel-search-based unmixing method. Although VCA is used to initialize the decoder weights corresponding to the endmembers, it only provides a relatively reasonable optimization starting point and helps accelerate convergence. The initialized weights are not fixed during training. Instead, they are treated as learnable parameters and are further optimized in an end-to-end manner. Therefore, the final estimated endmembers are not directly determined by VCA but are refined through network optimization.

Considering that an explicit volume constraint can restrict the simplex spanned by the estimated endmembers and provide additional geometric regularization, we further conduct an ablation study on the volume loss. Given the estimated endmember matrix $\hat{\boldsymbol{E}} = [\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2, \ldots, \hat{\boldsymbol{e}}_P] \in \mathbb{R}^{B\times P}$, we select $\hat{\boldsymbol{e}}_P$ as the reference vertex and construct the translated edge-vector matrix as $\tilde{\boldsymbol{E}} = [\hat{\boldsymbol{e}}_1 - \hat{\boldsymbol{e}}_P, \hat{\boldsymbol{e}}_2 - \hat{\boldsymbol{e}}_P, \ldots, \hat{\boldsymbol{e}}_{P-1} - \hat{\boldsymbol{e}}_P] \in \mathbb{R}^{B\times(P-1)}$. According to the high-dimensional simplex volume formula, the simplex volume can be computed from the Gram determinant of the edge-vector matrix [57], [58]. Therefore, the volume loss is defined as

$$L_{\mathrm{vol}} = \sqrt{\det\left(\tilde{\boldsymbol{E}}^T\tilde{\boldsymbol{E}} + \epsilon \boldsymbol{U}_{P-1}\right)} \tag{21}$$

where $\det(\cdot)$ denotes the determinant operator, ϵ is a small positive constant used to improve numerical stability, and $\boldsymbol{U}_{P-1}$ denotes the $(P-1)$-dimensional identity matrix. Since the constant scaling factor in the standard simplex-volume formula does not affect the optimization process, it is omitted in the loss function. After introducing the volume loss, the overall training objective becomes

$$L = \alpha L_{\mathrm{RE}} + \beta L_{\mathrm{SAD}} + \lambda L_{\mathrm{vol}}. \tag{22}$$

Table V reports the unmixing results of MDFPF-Net with and without the volume loss. It can be observed that both settings achieve favorable unmixing performance. After introducing the volume loss, some metrics can be further improved. For example, on the synthetic dataset, the aRMSE decreases from 0.0594 to 0.0551. These results indicate that the volume loss can be conveniently incorporated into MDFPF-Net as an

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TABLE V
COMPARISON OF UNMIXING PERFORMANCE WITH AND WITHOUT VOLUME LOSS

Dataset	Without vol. loss		With vol. loss	
	aRMSE	aSAD	aRMSE	aSAD
Synthetic (20 dB)	0.0594	0.0574	0.0551	0.0716
Samson	0.0507	0.0393	0.0544	0.0437
Jasper Ridge	0.0711	0.0539	0.0707	0.0631
Apex	0.1113	0.0813	0.1159	0.0782
Orchard	0.1019	0.1014	0.1158	0.0892

TABLE VI
COMPUTATIONAL TIMES (IN SECONDS) FOR DIFFERENT DATASETS

Methods	Synthetic	Samson	Jasper	Apex	Orchard	Cuprite
VCA-FCLS [14]	0.44	0.63	0.71	0.76	0.34	1.56
SGSNMF [17]	3.66	7.58	11.03	13.32	2.77	39.46
DAEN [26]	18.46	15.87	10.52	39.36	11.73	70.75
CyCU [28]	13.22	15.94	16.32	23.41	15.46	38.17
MiSiC [31]	18.49	9.41	11.14	12.62	20.73	74.20
DeepTrans [36]	15.65	14.51	17.33	19.81	26.33	89.37
A2SAN [34]	48.12	32.63	87.51	120.34	57.11	167.31
MAT [38]	19.21	14.51	34.14	40.76	22.38	82.74
SSAF [35]	44.65	45.91	104.19	92.95	44.06	136.87
UNMamba [39]	48.22	36.29	44.47	52.11	27.33	123.76
Proposed	35.94	12.78	14.54	36.06	40.71	102.56

auxiliary loss term, imposing an additional constraint on the simplex formed by the estimated endmembers at the loss-function level. Moreover, for the model without the volume loss term, multidomain feature representation, reconstruction consistency, spectral angle consistency, as well as the abundance nonnegativity and sum-to-one constraints enforced by the Softmax function jointly constrain the optimization process and reduce the uncertainty of endmember and abundance combinations, thereby guiding the model to learn reasonable and stable endmember and abundance estimations. Therefore, the ablation study on the volume loss not only demonstrates that MDFPF-Net can flexibly incorporate explicit volume regularization but also further supports the practical effectiveness of the proposed model in no-pure-pixel scenarios.

C. Running Time

Table VI reports the computational time of different methods on each dataset. All experiments were conducted on a workstation equipped with an Intel Core i7-12700H CPU, 16-GB RAM, and an NVIDIA GeForce RTX 4060 GPU. As the data complexity increases, most methods show a noticeable increase in computational time. Traditional methods usually require less time because they do not involve deep feature learning or complex network structures. It is worth noting that MDFPF-Net does not achieve the lowest computational time on all datasets. For example, on the synthetic dataset, MDFPF-Net takes 35.94 s, which is higher than several lightweight DL-based methods, such as DAEN, CyCU, and MiSiC. This is mainly because the proposed model introduces the FMBA module, CDDP module, and Transformer encoder to perform multidomain feature extraction and adaptive cross-domain fusion, which inevitably increases the computational cost. Nevertheless, compared with more complex attention- or Transformer-based unmixing methods, such as A2SAN and SSAF, MDFPF-Net still shows competitive runtime efficiency on several real-world datasets, especially Jasper Ridge and Apex, while maintaining strong unmixing accuracy in terms of RMSE and SAD. Therefore, the proposed method achieves a favorable balance between unmixing accuracy and computational efficiency.

V. CONCLUSION

This article introduces MDFPF-Net, a multidomain feature fusion model for HU based on an AE architecture. The core characteristics of this network include the use of a dual-branch CNN extractor to capture spatial–spectral features, the design of an FMBA mechanism to extract frequency-domain features, and the utilization of a multikernel dynamic perception module to achieve adaptive cross-domain semantic alignment and complementary enhancement among multidomain features. Furthermore, the model combines a Transformer encoder with a decoder equipped with the SEAG module, thereby effectively utilizing the fused features for accurate unmixing. Comparative experimental results on multiple datasets show that MDFPF-Net generally outperforms existing comparison methods in terms of RMSE and SAD, verifying the effectiveness of multidomain feature extraction and adaptive fusion in improving endmember discrimination and abundance estimation accuracy. Overall, the proposed method introduces frequency-domain feature modeling and an adaptive fusion mechanism into HU, which not only improves unmixing performance but also provides an effective multidomain feature representation and fusion strategy for HU in complex mixed scenes. Considering the complexity of endmember distributions and mixing mechanisms in real hyperspectral scenes, future research may further focus on fine-grained and lightweight fusion of multiple features, the introduction of multimodal auxiliary data such as light; detection and ranging image (LiDAR) and synthetic aperture radar image (SAR), and the modeling of complex nonlinear mixing effects, so as to further improve unmixing accuracy and robustness.

REFERENCES

[1] C. Zhu, S. Deng, X. Song, Y. Li, and Q. Wang, "Mamba collaborative implicit neural representation for hyperspectral and multispectral remote sensing image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504915.

[2] M. Wang et al., "Inversion of water quality parameters from UAV hyperspectral data based on intelligent algorithm optimized backpropagation neural networks of a small rural river," *Remote Sens.*, vol. 17, no. 1, p. 119, Jan. 2025.

[3] L. Hu, W. He, L. Zhang, and H. Zhang, "TeDFL: Toward text-driven few-shot learning for forest tree species classification with airborne hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5530115.

[4] J. Habashi, H. Jamshid Moghadam, M. Mohammady Oskouei, A. B. Pour, and M. Hashim, "PRISMA hyperspectral remote sensing data for mapping alteration minerals in SAR-e-châh-e-shur region, birjand, Iran," *Remote Sens.*, vol. 16, no. 7, p. 1277, Apr. 2024.

[5] Y. Xiang et al., "DFFormer: Capturing dynamic frequency features to locate image manipulation through adaptive frequency transformer and prototype learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 2, pp. 1907–1919, Feb. 2026.

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[6] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[7] L. Sun, H. Zhang, Y. Zheng, Z. Wu, Z. Ye, and H. Zhao, "MASSFormer: Memory-augmented spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516415.

[8] X. Zhang et al., "RSVMamba for tree species classification using UAV RGB remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5607716.

[9] L. Sun et al., "Local–global information perception network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5625518.

[10] Y. Liu, Q. Ye, L. Sun, and Z. Wu, "SOD-YOLOv8n: Small object detection in remote sensing images based on YOLOv8n," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[11] C. Li, T. Sun, K. F. Kelly, and Y. Zhang, "A compressive sensing and unmixing scheme for hyperspectral data processing," *IEEE Trans. Image Process.*, vol. 21, no. 3, pp. 1200–1210, Mar. 2012.

[12] M. E. Winter, "*N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data*," in *Imaging Spectrometry V*, vol. 3753. Bellingham, WA, USA: SPIE, 1999, pp. 266–275.

[13] J. W. Boardman, F. A. Kruse, and R. O. Green, "Mapping target signatures via partial unmixing of AVIRIS data," in *Proc. 5th Annu. JPL Airborne Earth Sci. Workshop (AVIRIS Workshop)*, vol. 1, 1995, p. 176.

[14] J. M. P. Nascimento and J. M. B. Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[15] D. C. Heinz and Chein-I-Chang, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.

[16] L. Miao and H. Qi, "Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 3, pp. 765–777, Mar. 2007.

[17] X. Wang, Y. Zhong, L. Zhang, and Y. Xu, "Spatial group sparsity regularized nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6287–6304, Nov. 2017.

[18] S. Zhang, J. Li, K. Liu, C. Deng, L. Liu, and A. Plaza, "Hyperspectral unmixing based on local collaborative sparse regression," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 631–635, May 2016.

[19] S. Zhang, J. Li, H.-C. Li, C. Deng, and A. Plaza, "Spectral–spatial weighted sparse regression for hyperspectral image unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3265–3276, Jun. 2018.

[20] L. Sun, F. Wu, T. Zhan, W. Liu, J. Wang, and B. Jeon, "Weighted nonlocal low-rank tensor decomposition method for sparse unmixing of hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1174–1188, 2020.

[21] L. Sun, F. Wu, C. He, T. Zhan, W. Liu, and D. Zhang, "Weighted collaborative sparse and $L_{1/2}$ low-rank regularizations with superpixel segmentation for hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 5500405.

[22] B. Rasti, B. Koirala, P. Scheunders, and P. Ghamisi, "UnDIP: Hyperspectral unmixing using deep image prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504615.

[23] B. Rasti, A. Zouaoui, J. Mairal, and J. Chanussot, "Image processing and machine learning for hyperspectral unmixing: An overview and the HySUPP Python package," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517631.

[24] Y. Su, A. Marinoni, J. Li, J. Plaza, and P. Gamba, "Stacked nonnegative sparse autoencoders for robust hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 9, pp. 1427–1431, Sep. 2018.

[25] Y. Qu and H. Qi, "UDAS: An untied denoising autoencoder with sparsity for spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1698–1712, Mar. 2019.

[26] Y. Su, J. Li, A. Plaza, A. Marinoni, P. Gamba, and S. Chakravortty, "DAEN: Deep autoencoder networks for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4309–4321, Jul. 2019.

[27] B. Palsson, J. R. Sveinsson, and M. O. Ulfarsson, "Spectral–spatial hyperspectral unmixing using multitask learning," *IEEE Access*, vol. 7, pp. 148861–148872, 2019.

[28] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503914.

[29] X. Lu, T. Li, J. Zhang, and F. Jia, "A novel unmixing-based hypersharpening method via convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503614.

[30] M. Zhao, S. Shi, J. Chen, and N. Dobigeon, "A 3-D-CNN framework for hyperspectral unmixing with spectral variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521914.

[31] B. Rasti, B. Koirala, P. Scheunders, and J. Chanussot, "MiSiCNet: Minimum simplex convolutional network for deep hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522815.

[32] L. Qi, F. Gao, J. Dong, X. Gao, and Q. Du, "SSCU-Net: Spatial–spectral collaborative unmixing network for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5407515.

[33] S. Hu and H. Li, "Hyperspectral unmixing with multi-scale convolution attention network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 2531–2542, 2024.

[34] X. Tao, M. E. Paoletti, Z. Wu, J. M. Haut, P. Ren, and A. Plaza, "An abundance-guided attention network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505414.

[35] W. Gao, J. Yang, Y. Zhang, Y. Akoudad, and J. Chen, "SSAF-Net: A spatial–spectral adaptive fusion network for hyperspectral unmixing with endmember variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5506015.

[36] P. Ghosh, S. K. Roy, B. Koirala, B. Rasti, and P. Scheunders, "Hyperspectral unmixing using transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535116.

[37] Y. Duan, X. Xu, T. Li, B. Pan, and Z. Shi, "UnDAT: Double-aware transformer for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522012.

[38] P. Wang, R. Liu, and L. Zhang, "MAT-Net: Multiscale aggregation transformer network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538115.

[39] D. Chen, J. Zhang, and J. Li, "UNMamba: Cascaded spatial–spectral mamba for blind hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, 2025, Art. no. 5502405.

[40] M. Zhang, H. Xie, M. Yang, Q. Jiao, L. Xu, and X. Tan, "Mamba-enhanced spatial–spectral feature learning for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 22798–22815, 2025.

[41] C. Zhan, H. Liu, and J. Zhang, "Hyperspectral unmixing with autoencoder network in wavelet domain," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 3259–3262.

[42] H. Liu, Y. Lu, Z. Wu, Q. Du, J. Chanussot, and Z. Wei, "Spectral variability Bayesian unmixing for hyperspectral sequence in wavelet domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522719.

[43] M. Xu, J. Xu, S. Liu, H. Sheng, B. Shen, and K. Hou, "Stationary wavelet convolutional network with generative feature learning for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5501613.

[44] B. N. Patro, V. P. Namboodiri, and V. S. Agneeswaran, "SpectFormer: Frequency and attention is what you need in a vision transformer," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Feb. 2025, pp. 9543–9554.

[45] L. Chen, L. Gu, L. Li, C. Yan, and Y. Fu, "Frequency dynamic convolution for dense image prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 30178–30188.

[46] Y. Chen, X. Zhang, B. Zhang, S. Xiong, S. Xiong, and X. Lu, "A dual-stage wavelet and linear attention enhancement network for agricultural hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5533217.

[47] L. Chen, Y. Fu, L. Gu, C. Yan, T. Harada, and G. Huang, "Frequency-aware feature fusion for dense image prediction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 10763–10780, Dec. 2024.

[48] Q. Wang, Z. Li, S. Zhang, N. Chi, and Q. Dai, "WaveFusion: A novel wavelet vision transformer with saliency-guided enhancement for multimodal image fusion," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 8, pp. 7526–7542, Aug. 2025.

[49] Z. Ying et al., "AEGL-Net: Adaptive multiscale global–local feature fusion network for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5627219.

[50] L. Sun et al., "MDC-FusFormer: Multiscale deep cross-fusion transformer network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528914.

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[51] M. Lou and Y. Yu, "OverLoCK: An overview-first-look-closely-next ConvNet with context-mixing dynamic kernels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 128–138.
[52] F. Zhu, "Hyperspectral unmixing: Ground truth labeling, datasets, benchmark performances and survey," 2017, *arXiv:1708.05125*.
[53] F. Zhu, Y. Wang, S. Xiang, B. Fan, and C. Pan, "Structured sparse method for hyperspectral unmixing," *ISPRS J. Photogramm. Remote Sens.*, vol. 88, pp. 101–118, Feb. 2014.
[54] B. Somers, L. Tits, and P. Coppin, "Quantifying nonlinear spectral mixing in vegetated areas: Computer simulation model validation and first results," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 1956–1965, Jun. 2014.
[55] J. Stuckens, B. Somers, S. Delalieux, W. W. Verstraeten, and P. Coppin, "The impact of common assumptions on canopy radiative transfer simulations: A case study in citrus orchards," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 110, nos. 1–2, pp. 1–21, Jan. 2009. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0022407308001866
[56] L. Tits, B. Somers, J. Stuckens, and P. Coppin, "Validating nonlinear mixing models: Benchmark datasets from vegetated areas," in *Proc. 6th Workshop Hyperspectral Image Signal Process.*, Evol. Remote Sens. (WHISPERS), Jun. 2014, pp. 1–4.
[57] K. Sun, X. Geng, P. Wang, and Y. Zhao, "A fast endmember extraction algorithm based on Gram determinant," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 6, pp. 1124–1128, Jun. 2014.
[58] A. M. S. Ang and N. Gillis, "Algorithms and comparisons of nonnegative matrix factorizations with volume regularization for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4843–4853, Dec. 2019.



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